Deepfake Attacks & AI-Based Detection Methods

IE2092 – Introduction to Cyber Security

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Kuyilini. T
IT24101551
Department of Computer Systems Engineering

OVERVIEW

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Abstract

What are Deepfakes?

AI-generated synthetic media — videos, audio, images — that realistically depict people saying or doing things they never did.

Why Now?

Since 2017, deepfake technology has evolved from detectable face-swaps to near-perfect diffusion-model outputs that fool even experts.

This Report Covers

How deepfakes work, their full chronological evolution, four AI-based detection methods, and six future mitigation directions.

Key Concern

Deepfakes are used in fraud, political disinformation, and non-consensual intimate imagery — 96% of online deepfakes fall in this last category.

INTRODUCTION

What Are Deepfakes?

Definition

"Deepfake" = "Deep learning" + "Fake"

AI can alter how a person looks, sounds, or acts in media — including:

- Face-swap videos
- Lip-sync audio edits
- Fully synthetic videos
- Cloned voices from short audio samples

Threat Areas

Cybersecurity

Social engineering, fake executive calls, bypassing face-auth systems

Politics

Fabricated statements by politicians; election interference

Personal Safety

96% of online deepfakes are non-consensual intimate images

Finance

$25M lost in a single HK deepfake CFO video call (2023)

EVOLUTION

Evolution of Deepfake Technology

2017–18

Phase 1

GAN face-swaps on Reddit. FaceSwap & DeepFaceLab released as open-source.

2019–20

Phase 2

FOMM animates still photos. Wav2Lip syncs audio. FaceForensics++ benchmark.

2021–22

Phase 3

DeepFaceLive enables real-time fakes. Stable Diffusion bypasses GAN detectors.

2023+

Phase 4

LLM pipelines create full fake broadcasts. Adversarial attacks defeat detectors.

DETECTION METHODS

AI-Based Detection Methods

CNN Spatial Analysis

~95%

Detects pixel-level artefacts in individual frames — blending at face edges, skin texture anomalies.

LSTM Temporal Analysis

~90%

Checks biological timing — blinking, lip sync, head movement across video frames.

Vision Transformer (ViT)

~97%

Global attention across the full face catches lighting, shadow and texture mismatches CNNs miss.

Frequency Domain (FFT)

~88%

Detects invisible GAN fingerprints in image frequency data left by the upsampling process.

Biometric rPPG

~85%

Analyses subtle heartbeat-driven colour changes in facial skin — AI faces cannot reproduce this.

Multimodal Fusion

~98%

Combines all methods into one unified system — highest accuracy, highest complexity.

FUTURE DEVELOPMENTS

Six Key Future Developments

1

Blockchain Provenance (C2PA)

2025–27

Cryptographic fingerprint embedded at capture; any edit breaks it.

2

Federated Learning

2026–28

Train detectors across orgs without sharing private video data.

3

Real-Time On-Device Detection

2025–27

NPU chips in smartphones run detection on live video calls.

4

Adversarially Hardened Detectors

Ongoing

Ensemble models resist invisible adversarial perturbations.

5

Universal AI Watermarking

2026–28

AI tools embed invisible watermarks (e.g. Google SynthID) at creation.

6

Legal & Regulatory Frameworks

2026–30

EU AI Act, DEEPFAKES Accountability Act mandate labelling & penalties.

CONCLUSION

Conclusion & Key Takeaways

Rapid Evolution

Deepfakes evolved from 2017 GAN tools to 2025 foundation-model pipelines creating full fake broadcasts.

Arms-Race Dynamic

Detection always lags one step behind generation — a fundamental challenge with no easy solution.

Best Detection

Multimodal fusion (spatial + temporal + frequency + rPPG) achieves ~98% accuracy on benchmarks.

Multi-Layered Solution

No single fix works; effective mitigation needs technical, regulatory & public awareness efforts together.

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Full reference list available in the submitted report (22 IEEE references)

Thank You

Questions & Discussion

Kuyilini. T | IT24101551
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